

Bingi Gnaneswanth

✉ aswanthbingi@gmail.com ☎ +91 7013264835 ↗ LeetCode 🔗 LinkedIn 🐙 Github

SUMMARY

Computer Science Engineering student with hands-on experience in Java, Spring Boot, Python, SQL, backend systems, and machine learning. Strong foundation in data structures, algorithms, software engineering, and distributed systems, with experience building REST APIs, fault-tolerant systems, and data-driven decision systems using contextual bandits and optimization.

SKILLS

Languages: Java,C,Python,SQL

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Backend: Spring Boot, Spring MVC, REST APIs, gRPC, Protocol Buffers, Node.js

Frontend: React, TypeScript, TanStack Router, Tailwind CSS, HTML, CSS, D3.js

Databases: MySQL, PostgreSQL, MongoDB, RocksDB

AI/ML: Contextual Bandits, LinUCB, Online Learning, Off-Policy Evaluation, Policy Optimization, Machine Learning, RAG, FAISS, Sentence Transformers, ChromaDB

Tools: Git, Maven, Docker, Prometheus, Grafana

PROJECTS

Distributed Fault-Tolerant Key-Value Store using the Raft Consensus Algorithm

- Built a **fault-tolerant distributed key-value store** implementing Raft **leader election, log replication, quorum-based consensus, and automatic failover** for strong consistency.
- Implemented **RocksDB persistence, gRPC-based inter-node communication, crash recovery, and snapshotting** for reliable distributed state management.
- Developed a **React real-time cluster dashboard** and instrumented **25+ Prometheus metrics** for node health, consensus events, replication latency, RPC performance, and KV operations.
- **Containerized the multi-node cluster with Docker** and integrated **Prometheus/Grafana** for real-time observability, performance monitoring, and failure analysis.

Budget-Constrained Payment Recovery using Contextual Bandits

- Built a contextual-bandit recovery engine (**LinUCB**) that selects among six payment-recovery actions per failed transaction from transaction, failure-reason, and customer-behavior context — reframing one-off retry decisions as a **budget-constrained portfolio-allocation** problem.
- Designed **primal-dual budget pacing** (a learned shadow price) to allocate a shared ₹300/day recovery budget to the highest-value failures; beat a fixed-retry baseline in **all 10 randomized simulations** — **~40% higher net revenue on average (up to 50%)**, capturing ~84% of a budget-matched optimum.
- Built a **leakage-safe offline-evaluation framework** — time-based splits, fixed/random/oracle baselines, off-policy estimators (IPS/SNIPS/**Doubly-Robust**) with bootstrap confidence intervals, budget sweeps, and ablations — tracking recovery rate, regret, budget utilization, and net recovery.

EDUCATION

Dayananda Sagar University

Bachelor of Technology in Computer Science Engineering

Sep 2023 – Ongoing 2027 | Bangalore

Gitam Junior College

MPC-Mathematics, Physics, Chemistry

Jun 2022 – Apr 2023 | Tirupati